ADR-0001: Sealed quotes and a single-transaction confirmation

Status	Date	Decision makers	Confidence
accepted	2026-10-15	Tech Lead	High. Medium on gift tax

Confirmed orders went missing in three releases. We must guarantee confirmed total = quoted total, exact money, that a 201 is a stored order, and that effects on push and the ERP are never lost or applied twice.

Decisions and discarded alternatives

Decision	Discarded	Why
Sealed, insert-only quote. The order references it, with no amounts	Copying the breakdown. Recomputing on confirm	Copies diverge. Prices change
BigDecimal rounded per line, totals never re-rounded. Combos before tiers	long minor units. Rounding the total. Tiers first	Awkward with tax rates. Invoice mismatch. Double discount
One PostgreSQL ACID transaction, locks key → quote → credit. The 201 after commit	In-memory store. Asynchronous write or saga. Keys in a cache	No real locks. The "confirmed but missing" defect. Not atomic
Transactional outbox with lease and backoff. Push and ERP deduplicate by event_id	Publishing after commit. Two-phase commit	Lost events. Push and ERP cannot join it
Hexagonal monolith, explicit SQL, separate repositories	Microservices. JPA. Monorepo	One transaction. JPA hides locks. Components evolve apart

Not implemented, on purpose

- **Authentication:** out of scope. A gateway would inject tenant and customer.
- **Real push and ERP clients:** fakes behind ports prove the guarantees.
- **Not asked for:** revocation (the risk mitigation), cancellation, credit release, catalog, tax categories, price history, OpenAPI, deployment, CI and SonarQube (README).

Code review checklist

- No float or double. Money compared with compareTo. No UPDATE on quotes.
- @Transactional only on use cases, with no self-invocation and no remote calls.
- Locks key → quote → credit. The 201 after commit. No cached failure.
- The relay sends outside any transaction. Only the lease holder marks the outcome.
- Every lever with a golden test seen failing. User text only in i18n.

Risk and mitigation

Risk: a sealed quote honors a wrong price until it expires. **Mitigation:** a 15-minute TTL per country; then revocation (OPEN → REVOKED, 409). **Revisit** if an incident affects more than 100 quotes or the TTL exceeds one hour.